

Alessio Santoru

Senior Application Security Engineer

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PROFILE

Application security engineer with 8+ years spanning offensive security and secure-by-design — from penetration-testing consultancy to securing AWS infrastructure at global scale. Focused on building scalable security programs and automation, threat modeling complex systems, and vulnerability research. Experienced across the full lifecycle: assessing, securing, and hardening cloud-based systems and services.

EXPERIENCE

Senior Application Security Engineer

Dec 2022 – Present

Amazon Web Services (AWS), Milan

- › Led the Application Security program for AWS Outposts and Local Zones, enabling the secure launch of 50+ features.
- › Defined a threat model and enforced security requirements for operating AWS services in non-trusted environments, developing automation tools to scale security verifications across 20+ core AWS services.
- › Owned the Security Review process for EC2 VPC security and drove initiatives to identify and mitigate 10+ authorization anti-patterns and logical bypasses.

Senior Product Security Engineer

Oct 2021 – Oct 2022

Trade Republic, Berlin

- › Designed and implemented a comprehensive Secure Software Development Lifecycle (SSDLC) adopted by 10+ services, including automated security checks through custom-built internal platforms.
- › Developed threat models and risk prioritization frameworks per service, partnering closely with engineering teams to guide the implementation of effective security controls.
- › Conducted threat research to uncover emerging attack vectors and strengthen defenses against supply chain risks in third-party integrations.

Product Security Engineer

Nov 2019 – Sep 2021

N26, Berlin

- › Strengthened the security of mobile, web, and API ecosystems through expanded threat modeling and penetration testing coverage.
- › Built internal automation tools to streamline security testing across mobile release pipelines.
- › Performed vulnerability research and third-party software penetration testing to identify and mitigate risks.
- › Launched the Security Champions Program, training 30+ engineers across 5+ teams on Secure Development, Threat Modeling, and the OWASP Top 10.

Security Consultant / Penetration Tester

Apr 2017 – Oct 2019

Horizon Security, Milan

- › Conducted penetration tests and vulnerability assessments across web, mobile, hardware, and IoT environments for leading Italian companies in banking, insurance, automotive, betting, and energy.
- › Researched and reported vulnerabilities through responsible disclosure programs to 5+ third-party vendors.

EDUCATION

BSc Computer Science

Università degli Studi di Perugia
2017 • Final mark 110/110

CERTIFICATIONS

GXPN 2026

GIAC Exploit Researcher & Adv. Pen Tester

OSCP 2018

Offensive Security Certified Professional

LANGUAGES

Italian

Native speaker

English

Professional working proficiency

ACHIEVEMENTS / CVE

CVE-2022-2903 PHP Object Injection (WordPress NinjaForms) • **CVE-2018-20122** RCE (Fastweb FASTgate) • **CVE-2018-17172** Command Injection (Xerox AltaLink) • **CVE-2018-7064** Reflected XSS (Aruba Instant) • **CVE-2017-17663** Buffer Overflow (mini_httpd/thttpd)